

LANGCHAIN AI AGENT DEVELOPMENT REPORT

Document Q&A Integration with Deep Agents & Gemini 2.5 Flash

Date: July 1, 2026	Framework: LangChain / DeepAgents
Author: Developer (AI Assistant)	Target Model: google_genai:gemini-2.5-flash
Workspace: /Desktop/langchain	Status: Executed Successfully

1. Project Overview & Objective

This project implements a local **Document Question Answering (Q&A) system** using the deepagents wrapper around LangChain components. The agent is configured to ingest a demographic profile document, extract its content, and answer specific analytical queries. The objective is to perform *single-shot context injection* where the model responds strictly based on the extracted document content rather than pre-trained general knowledge.

2. System Architecture & File Structure

The workspace consists of a simple yet robust directory structure containing environment configurations, source document, and the main invocation script:

- **app.py:** The main entry point script containing logic for PDF text extraction and deep agent execution.
- **.env:** Configuration file holding the Google API Key (GOOGLE_API_KEY) required for GenAI authentication.
- **FF2023-28-Phil-Demographic-Profile.pdf:** The target source document containing demographic statistics of the Philippines.
- **.venv/:** The python virtual environment containing dependencies: google-genai, langchain, pypdf, and deepagents.

Workflow Pipeline:

1. **Text Extraction:** app.py uses pypdf.PdfReader to parse the local demographic document page-by-page and concatenate the raw text.
2. **Agent Initialization:** A deep agent is instantiated with a system prompt setting its role as a precise data analyst.
3. **Context Injection & Query:** The extracted text (6,259 characters) is dynamically wrapped inside a prompt template along with the user question.
4. **Inference:** The prompt is sent to Gemini, and the structured response is output to the console.

3. Source Code Implementation (app.py)

Below is the complete implementation of the LangChain agent script. It reads the local file, loads the agent, and executes the question-answering workflow:

```
from dotenv import load_dotenv
import os
load_dotenv()

from deepagents import create_deep_agent
from pypdf import PdfReader

# 1. READ THE PDF LOCAL FIRST
pdf_filename = "FF2023-28-Phil-Demographic-Profile.pdf"
pdf_text = ""

print("■ Reading the PDF file locally...")
try:
    reader = PdfReader(pdf_filename)
    for page in reader.pages:
        text = page.extract_text()
        if text:
            pdf_text += text + "\n"

    print(f"■ Successfully read {len(pdf_text)} characters from the document.")
except Exception as e:
    print(f"■ Error reading file: {e}")

# 2. CREATE A SIMPLE AGENT (NO TOOLS NEEDED)
agent = create_deep_agent(
    model="google_genai:gemin-2.5-flash",
    system_prompt="You are a precise data analyst. Answer the user's question using ONLY the provided document text context.",
)

# 3. BUILD THE SINGLE-SHOT INJECTION PROMPT
user_question = "What is the population of the Philippines according to the document?"

full_prompt = f"""
Below is the complete text extracted from the demographic profile document:
-----
{pdf_text}
-----

Based strictly on the text provided above, please answer this question: {user_question}
"""

print("■ Sending the text context and your question to Gemini in a single hop...")

# 4. EXECUTE THE INVOCATION
result = agent.invoke(
    {"messages": [{"role": "user", "content": full_prompt}]}
)

# 5. PRINT THE CLEAN RESPONSE DIRECTLY
final_message = result['messages'][-1]
print("\n■ Final Agent Response:")
print(final_message.content)
```

4. Debugging & Model Selection

During initial development and execution, the system encountered API communication errors. Below is a summary of the diagnostic history and resolution:

Model Error (404 Not Found):

The agent was initially set to use the gemini-1.5-flash model. However, invocation returned a client error: *'models/gemini-1.5-flash is not found for API version v1beta, or is not supported for generateContent.'*

Resolution:

The model parameter in create_deep_agent was updated to **google_genai:gemini-2.5-flash**. This matched the supported Google GenAI API endpoint, enabling successful authentication and inference.

5. Execution Results & Agent Output

When successfully executed, the terminal logs the reading status, character length, and the final response from the Gemini model:

[Reading] Reading the PDF file locally...

[Success] Successfully read 6259 characters from the document.

[Sending] Sending the text context and your question to Gemini in a single hop...

[Agent Response] Final Agent Response:

The Philippine total population reached 109,035,343 as of 01 May 2020.